

Calculus II Lesson 13

1) $\sum_{n=1}^{\infty} \frac{1}{n+7}$

a) Find $\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \frac{1}{n+7} =$ _____

b) What does the nth Term Test say about the series $\sum_{n=1}^{\infty} \frac{1}{n+7}$?

c) $\int_1^{\infty} \frac{1}{x+7} dx =$ _____ Hint: $\int \frac{1}{x+a} dx = \ln|x+a| + C$

d) $\sum_{n=1}^{\infty} \frac{1}{n+7} =$ _____ (finite value or ∞ ?)

e) Use the Integral Test to explain why the series $\sum_{n=1}^{\infty} \frac{1}{n+7}$ diverges.

2) $\sum_{n=1}^{\infty} \frac{1}{4^n}$?

a) Find $\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \frac{1}{4^n} =$ _____

b) What does the nth Term Test say about the series $\sum_{n=1}^{\infty} \frac{1}{4^n}$?

c) $\int_1^{\infty} \frac{1}{4^x} dx = \int_1^{\infty} \left(\frac{1}{4}\right)^x dx = ?$ Hint: $\int a^x dx = \frac{1}{\ln a} a^x + C$

d) $\sum_{n=1}^{\infty} \frac{1}{4^n} =$ _____ (finite value or ∞ ?)

e) Use the Integral Test to explain why the series $\sum_{n=1}^{\infty} \frac{1}{4^n}$ converges.

$$3) \sum_{n=1}^{\infty} e^{-n} = \sum_{n=1}^{\infty} \frac{1}{e^n} = \sum_{n=1}^{\infty} \left(\frac{1}{e}\right)^n$$

a) Find $\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} e^{-n} = \lim_{n \rightarrow \infty} \frac{1}{e^n} = \underline{\hspace{2cm}}$

b) What does the nth Term Test say about the series $\sum_{n=1}^{\infty} e^{-n}$?

c) $\int_1^{\infty} e^{-x} dx = ?$ Hint: $\int e^u du = e^u + C$

d) $\sum_{n=1}^{\infty} e^{-n} = \underline{\hspace{2cm}}$ (finite value or ∞ ?)

e) Use the Integral Test to explain why the series $\sum_{n=1}^{\infty} e^{-n}$ converges.

4) $\sum_{n=1}^{\infty} \frac{1}{n^2 + 1}$

a) Find $\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \frac{1}{n^2 + 1} =$ _____

b) What does the nth Term Test say about the series $\sum_{n=1}^{\infty} \frac{1}{n^2 + 1}$?

c) $\int_1^{\infty} \frac{1}{x^2 + 1} dx =$ _____ Hint: Use Table of Integrals.

d) $\sum_{n=1}^{\infty} \frac{1}{n^2 + 1} =$ _____ (finite value or ∞ ?)

e) Use the Integral Test to explain why the series $\sum_{n=1}^{\infty} \frac{1}{n^2 + 1}$ converges.

5) $\sum_{n=2}^{\infty} \frac{\ln n}{n}$

a) Find $\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \frac{\ln n}{n} = \underline{\hspace{2cm}}$

b) What does the nth Term Test say about the series $\sum_{n=2}^{\infty} \frac{\ln n}{n}$?

c) $\int_2^{\infty} \frac{\ln x}{x} dx = \int_2^{\infty} \ln x \cdot \left(\frac{1}{x}\right) dx = ?$ Hint: Let $u = \ln x$

d) $\sum_{n=2}^{\infty} \frac{\ln n}{n} = \underline{\hspace{2cm}}$ (finite value or ∞ ?)

e) Use the Integral Test to explain why the series $\sum_{n=2}^{\infty} \frac{\ln n}{n}$ diverges .

6) $\sum_{n=1}^{\infty} \frac{\ln n}{n^2}$

a) Find $\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \frac{\ln n}{n^2} = \underline{\hspace{2cm}}$

b) What does the nth Term Test say about the series $\sum_{n=1}^{\infty} \frac{\ln n}{n^2}$?

c) $\int_1^{\infty} \frac{\ln x}{x^2} dx = ?$ Hint: $\int \frac{\ln x}{x^2} dx = \frac{-(\ln x + 1)}{x}$

d) $\sum_{n=1}^{\infty} \frac{\ln n}{n^2} = \underline{\hspace{2cm}}$ (finite value or ∞ ?)

e) Use the Integral Test to explain why the series $\sum_{n=1}^{\infty} \frac{\ln n}{n^2}$ converges

7) $\sum_{n=1}^{\infty} \frac{1}{(2n+1)^4}$

a) Find $\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \frac{1}{(2n+1)^4} =$ _____

b) What does the nth Term Test say about the series $\sum_{n=1}^{\infty} \frac{1}{(2n+1)^4}$?

c) $\int_1^{\infty} \frac{1}{(2x+1)^4} dx = ?$ Hint: Let $u = 2x+1$

d) $\sum_{n=1}^{\infty} \frac{1}{(2n+1)^4} =$ _____ (finite value or ∞ ?)

e) Use the Integral Test to explain why the series $\sum_{n=1}^{\infty} \frac{1}{(2n+1)^4}$ converges.

8) $\sum_{n=1}^{\infty} \frac{n}{2n^2 + 1}$

a) Find $\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \frac{n}{2n^2 + 1} =$ _____

b) What does the nth Term Test say about the series $\sum_{n=1}^{\infty} \frac{n}{2n^2 + 1}$?

c) $\int_1^{\infty} \frac{x}{2x^2 + 1} dx = \int_1^{\infty} \frac{1}{2x^2 + 1} x dx = ?$ Hint: Let $u = 2x^2 + 1$

d) $\sum_{n=1}^{\infty} \frac{n}{2n^2 + 1} =$ _____ (finite value or ∞ ?)

e) Use the Integral Test to explain why the series $\sum_{n=1}^{\infty} \frac{n}{2n^2 + 1}$ diverges.

